SECTION-I(ii) : One or More than one

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4 (–1)**

1. Which of the natural amino acids are example of non-essential amino acids?

(A) Glutamic acid (B) Aspartic acid (C) Asparagine (D) Glutamine

2. If glutamic acid is placed in an electric field, its migration depend on pH of the solution.Which of the following options are correct regarding alanine?

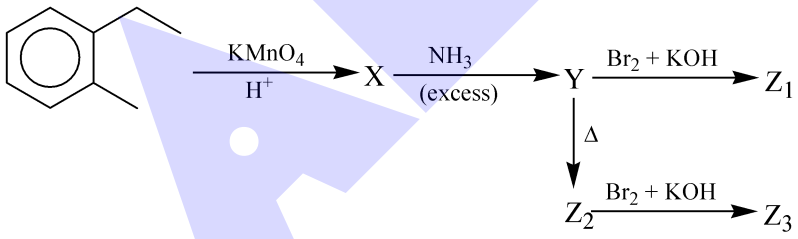
(A) At pH = 3.2, there is no net migration of alanine towards a particular electrode

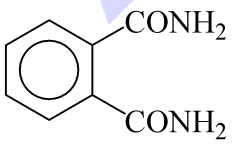
(B) At pH = 9.8, alanine migrates towards anode

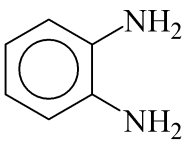
(C) At pH = 3.2, Alanine migrates towards cathode

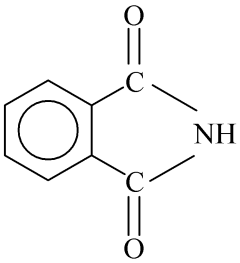
(D) At pH = 6, Alanine migrates towards anode

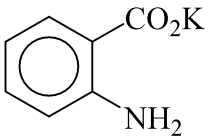
3. The final product formed is called phthalimide in the following reaction sequence. Which of the following statement is correct regarding product formed in each step of following reaction sequence.



(A) Y is 

(B) Z_1 is 

(C) Z_2 is 

(D) Z_3 is 

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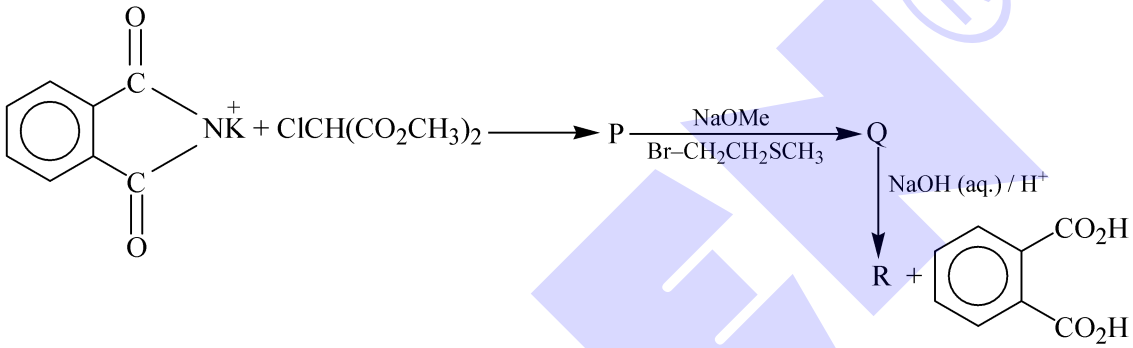
4. Which of the following polyers have vinylic monomer units?

(A) Acrilan (B) Polystyrene (C) Nylon (D) Teflon

SECTION–I(iii) : Paragraph Type

This section contains **3 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.No. 5 to 7

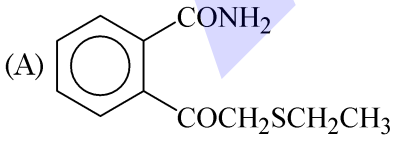


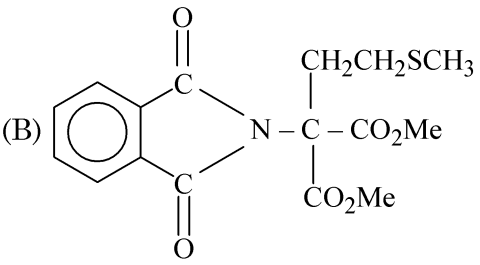
5. Compound R is :

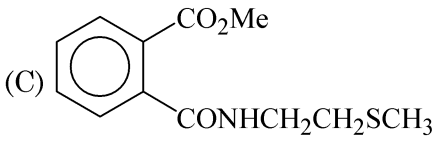
(A) $CH_3SCH_2CH(NH_2)CH_2CO_2H$ (B) $CH_3SCH(NH_2)CH_2CH_2CO_2H$

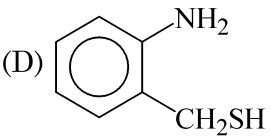
(C) $CH_3SCH_2CH_2CH(NH_2)CO_2H$ (D) $HSCH_2CHCH_2CH(NH_2)CO_2H$

6. Compound Q is :

(A) 

(B) 

(C) 

(D) 

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7. P to Q formation is example of

(A) Acid–base reaction followed by elimination reaction

(B) Nucleophilic substitution reaction followed by elimination reaction

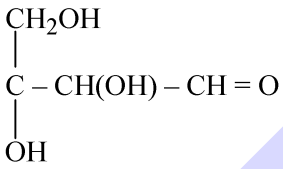
(C) Elimination reaction followed by Nucleophilic substitution reaction

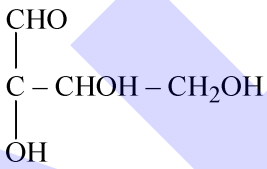
(D) Acid-base reaction followed by Nucleophilic substitution reaction

Paragraph for Q.no. 8 to 10

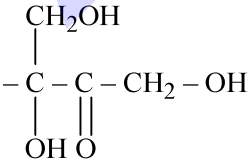
Compound A is example of aldose with molecular formula $C_5H_{10}O_5$. It gives a tetra-acetate with Ac_2O. A reacts with Br_2 / $H_2O$to give acid B with molecular formula $C_5H_{10}O_6$. Reduction of A with HI gives isopentane.

8. The structure of A can be

(A) 

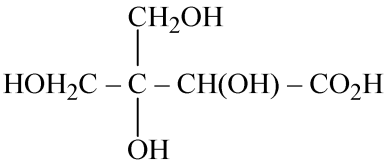
(B) 

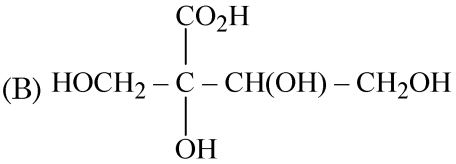
(C) Both (A) and (B)

(D) 

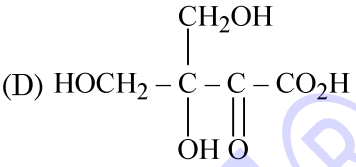
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9. The structure of B can be –

(A) 

(B) 

(C) Both (A) and (B)

(D) 

10. A can have

(A) 3° alcoholic –OH group

(B) 1° alcoholic –OH group

(C) 2° alcoholic –OH group

(D) All

Paragraph for Q.no. 11 to 13

	Column-I (Polymer)		**Column-II (Monomer)**		**Column-III (Properties of monomer)**
(I)	Polystyrene	(i)	Caprolactum	(P)	Resonance stabilization
(II)	Poly Vinyl chloride	(ii)	$PhCH = CH_2$	(Q)	1,4 addition of HBr (mainly) at 40°C
(III)	Natural rubber	(iii)	$CH_2 = CHCl$	(R)	On addition of HCl gemdichloride is formed
(IV)	Nylon-6	(iv)	$CH_2=C(CH_3)–CH=CH_2$	(S)	CO_2(g) evolution with $KMnO_4/H^+$

11. Which of the following combination is correct?

(A) (III), (ii), (R) (B) (II), (i), (P) (C) (IV), (i), (P) (D) (I), (iii), (S)

12. Which of the following combination is incorrect?

(A) (I), (ii), (P) (B) (I), (ii), (S) (C) (III), (iv), (P) (D) (II), (iv), (R)

13. Which of the following combination is correct?

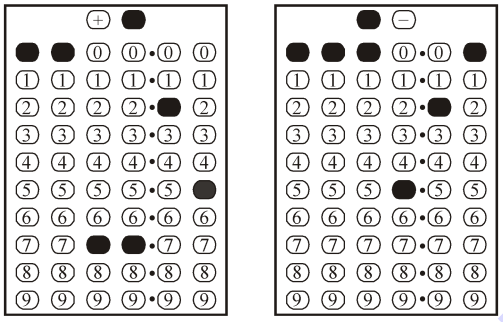
(A) (II), (ii), (P) (B) (II), (iii), (R) (C) (III), (iii), (P) (D) (IV), (ii), (P)

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SECTION-II : (Maximum Marks: 20)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



4(0)

1. What is the net charge of monosodium glutamate (MSG)?

2. Number of aldaric acids obtained from 16 aldohexose

3. A hexapeptide (m.wt. = 660) on complete hydrolysis gives glycine (m.wt. = 75) and two other amino acids. Glycine contributes 40% of total weight of hydrolysed products. The number of glycin unit present in hexapeptide is –

4. If molecular formula for a polypentoside (pentosan) is $[C_aH_bO_c]_n$ then find out the value of a + c – b?

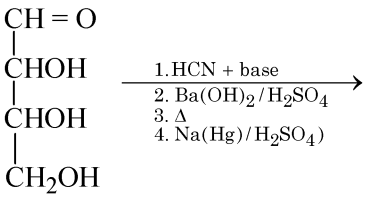
5. How many of the following compounds will give silver mirror with Tollen's reagent

MeCHO, PhCHO, PhNHOH, Propyne, $MeCO_2H$, HCO_2H, Sucrose, Starch, α-D-Glucopyranoside

6. The number of carbon atoms in smallest aldose able to for a cyclic stable hemiacetal

7. If x is the number of chiral centre in α–D–glucopyranose and y is the number of chiral center of β-D-fructofuranose then find out sum of x + y = ?

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8. 

Number of carbon atoms present in P

SECTION–IV : Matrix Match

This section contains **FIVE** questions. Each question contains two columns, **Column-I** and **Column-II. Column-I** has **four** entries (A), (B), (C) and (D) **Column-II** has **five** entries (P), (Q), (R), (S) and (T) Match the entries in **Column-I** with the entries in **column-II.** One or more entries in **Column-I** may match with one or more entries in **Column-II.** The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below : **4(0)**

(A) (P) (Q) (R) (S) (T)

(B) (P) (Q) (R) (S) (T)

(C) (P) (Q) (R) (S) (T)

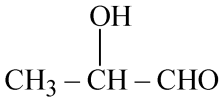
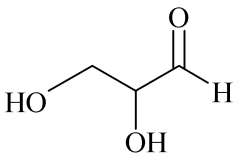
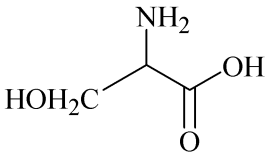
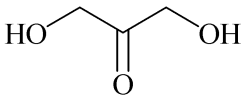
(D) (P) (Q) (R) (S) (T)

1.

	Column-I **Type of polypeptide**		**Column-II** **Feature**
(A)	Ala gly phenala	(P)	Tripeptide
(B)	Val Leu isoleu	(Q)	Tetrapeptide
(C)	Gln Gly Glu	(R)	On hydrolysis 2 chiral amino acids and one achiral amino acid are formed.
(D)	Ala Asn Asp Gly	(S)	On hydrolysis 3 neutral amino acids are formed
		(T)	Odd number of chiral centre is present

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2.

	Column-I **Structure of compounds**		**Column-II** **Type / test**
(A)		(P)	Carbohydrate
(B)		(Q)	Amino acid
(C)		(R)	Positive Tollen's test
(D)		(S)	Ninhydrin test
		(T)	Positive $NaHCO_3$ Test

3.

	Column-I		**Column-II**
(A)	Polymer of methyl methacrylate	(P)	Lexan
(B)	Polymer of 2-chloro 1,3-butadiene	(Q)	Kevlar
(C)	Polyamide	(R)	Plexiglass
(D)	Polycarbonate	(S)	Neoprene
		(T)	Synthetic polymer

4.

	Column-I		**Column-II**
(A)	Polyester of glycol and phthalic acid	(P)	Novolac
(B)	Copolymer of 1,3–butadiene and styrene	(Q)	Glyptal
(C)	Phenol and formaldehyde resin	(R)	Buna–S
(D)	Polyester of glycol and terephthalic acid	(S)	Dacron
		(T)	Co-polymer and condensation polymer

5.

	Column–I		**Column–II**
(A)	Guta–purcha	(P)	Chain growth polymer
(B)	Glyptal	(Q)	Step growth polymer
(C)	Nylon-2-nylon-6	(R)	Natural polymer
(D)	Cellulose	(S)	Biodegradable polymer
		(T)	Glycosidic linkage

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